

Learning C Through Small Projects

This document records my progress learning the C programming language through a structured series of small, hands-on projects.

Each lesson introduces one major concept and builds on the previous exercises.

Learning roadmap

Lesson	Main concept	Status
1	Compilation, Make, and <code>main()</code>	Completed
2	<code>argc</code> , <code>argv</code> , and exit codes	Completed
3	Variables and arithmetic	Completed
4	Loops and control flow	Completed
5	Functions	Next
6	Header files and modules	Planned
7	Strings	Planned
8	File input and output	Planned
9	Structures	Planned
10	Dynamic memory	Planned
11	Pointers and arrays	Planned
12	Mini Unix utility	Planned

Completed lessons

Lesson 1: Hello World

This lesson introduced the basic structure of a C program, including `#include`, the `main()` function, `printf()`, compilation with GCC, and automated builds using Make.

Lesson 2: Command-line arguments

This lesson introduced `argc` and `argv`, which allow a C program to receive arguments from the command line. It also introduced program exit codes and basic input validation.

Lesson 3: Simple calculator

This lesson introduced variables, numeric types, arithmetic operators, integer and floating-point division, and protection against division by zero.

Lesson 4: Loops and control flow

This lesson introduced:

- `for` loops
- `while` loops
- `if-else` conditional branching
- `break` to exit a loop
- `continue` to skip the remainder of an iteration

- return to exit the current function

The exercises demonstrated that statements execute sequentially. Control-flow statements only affect code that has not yet executed.

For example:

```
if (i == 5) {  
    continue;  
}  
  
printf("%d\n", i);
```

Equivalent Python exercises were created to compare universal programming concepts with language-specific syntax.

Comparative exercises

Selected C exercises are recreated in other languages when the comparison strengthens understanding.

Python provides comparison with a higher-level general-purpose language. PowerShell will provide comparison with scripting, structured objects, pipelines, and Windows automation.

C remains the primary learning track.

Current position

Lessons 1 through 4 are complete. The next lesson introduces functions and uses them to refactor the calculator from Lesson 3.